Graphic Era

End Term (Odd) Semester Examination December 2024

Roll no.....

Name of the Course and semester: B.TECH (ECE), V Semester
Name of the Paper: Object Oriented Programming using C++
Paper Code: TOE 501

Time: 3 hour

Maximum Marks: 100

Note:

(i) All the questions are compulsory.
(ii) Answer any two sub questions from a, b and c in each main question.
(iii) Total marks for each question is 20 (twenty).
(iv) Each sub-question carries 10 marks.

Q1. (2X10=20 Marks) (CO1)

a. Define friend functions in C++ and explain their concept. Illustrate with an example how friend functions can access private and protected members of a class.

b. Analyze the significance of encapsulation in C++ and evaluate how it ensures data protection and simplifies complex systems.

c. Explain the importance of static members in C++ classes. Differentiate between static data members and static member functions, and demonstrate their usage with examples.

Q2. (2X10=20 Marks) (CO2)

a. Discuss the role of constructors and destructors in managing the object lifecycle. Compare parameterized constructors with default constructors in terms of their usage.

b. Explain the concept of dynamic memory allocation in C++ and demonstrate how constructors handle dynamic memory allocation. Analyze the importance of releasing dynamically allocated memory using destructors, providing examples.

c. Evaluate the advantages and disadvantages of operator overloading in C++. Illustrate your answer by overloading an assignment operator with an example.

Q3. (2X10=20 Marks) (CO3)

a. Define templates in C++ and classify the different types of templates that can be used to create generic classes. Demonstrate each type with examples.

b. Explain exception specifications in C++, discuss their purpose, and illustrate how to specify exceptions a function may throw. Provide the syntax and an example.

c. Analyze uncaught exceptions in C++ and discuss how they are handled. Explain the role of the `throw()` exception specification in function declarations, with an example.

Q4. (2X10=20 Marks) (CO4)

a. Compare static polymorphism and dynamic polymorphism in C++. Illustrate with examples how static polymorphism is achieved through function overloading and templates, and how dynamic polymorphism is achieved through virtual functions.

b. Define the diamond problem in C++ multiple inheritance. Explain how virtual inheritance resolves this issue and illustrate the use of virtual base classes with an example that avoids ambiguity.

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c. Explain virtual destructors in C++ and analyze why destructors need to be virtual in a polymorphic class hierarchy. Demonstrate the consequences of not using a virtual destructor with an example.

Q5. (2X10=20 Marks) (CO5)

a. Explain random access in file handling and analyze how random access files enable reading and writing at specific positions in a file. Demonstrate the use of seekg and seekp functions with an example.

b. Discuss the importance of object serialization in C++ and illustrate the process of serializing an object to a file and deserializing it back into a program with an example.

c. Explain I/O manipulators in C++ and demonstrate their use with an example that formats output data with appropriate alignment, precision, and width.